

Can a Collective Think?

From Individual Intelligence to Collective Intelligence

*"Perhaps intelligence does not always belong to an individual.
It may sometimes belong to the relationships between individuals."*

— Monendra Grover (2026)

Different parts.
Shared information.
Emergent behaviour.
A wider view of life.

More than the sum
of its parts.

Is intelligence only in brains?

We usually associate intelligence with brains. But many forms of life – from bacteria to plants to immune systems – show remarkable, intelligent behaviour without a brain.



A new question

If intelligence can exist without a brain, can it exist without an individual?

What if intelligence is sometimes a property of a collective?



The key idea

The whole can know something that no part knows. Through interaction, communication and feedback, simple individuals can give rise to surprisingly sophisticated collective behaviour.

*"The intelligence is not in any single ant.
It emerges from the network."*



Examples of collective intelligence in biology

Ant colonies

Find efficient routes, defend nests, solve problems – without a central controller.



Microbial communities

Communicate, cooperate, compete and adapt together.



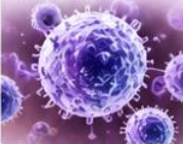
Plants

Sense their environment, communicate chemically and adjust their growth.



Immune system

Recognizes threats, remembers past encounters and coordinates responses across the body.



Animal groups

Bird flocks, fish schools and other groups show coordinated behaviour that no individual could achieve alone.



Ecosystems

Organisms and their environment exchange information in complex networks.

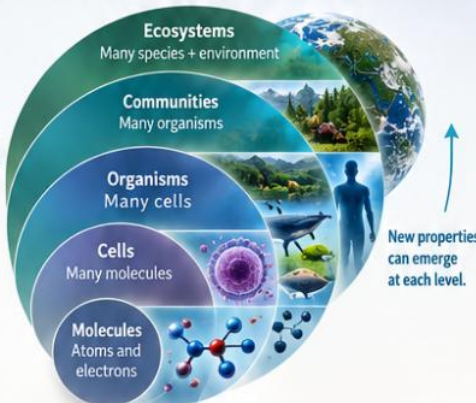


What does a collective do?

- Integrates information from many sources
- Distinguishes between different contexts
- Adapts based on experience
- Retains some form of memory
- Solves problems that individuals cannot solve alone
- Remains functional even when parts fail

These are some of the properties we usually associate with intelligence.

Levels of biological organization



Not every complex system is intelligent

Complexity alone is not enough. A hurricane or a crystal can be complex but they do not process information in the way living systems do.



A hurricane is complex, but not intelligent.



A crystal is ordered, but not intelligent.

Collective intelligence requires information integration, interaction, adaptation, memory and context-sensitive behaviour.

Lessons for artificial intelligence



Biology suggests another path. Intelligence may emerge from many simple units interacting, rather than a single powerful processor.

Perhaps the future of AI lies in intelligent collectives, not just bigger brains.

Rethinking individuality

We are not isolated entities. We are networks within networks – made of cells, microorganisms and interactions with the environment.

The boundary between individual and collective is more fluid than we imagine.



Evolution as a collective process



Evolution can be seen as a vast distributed search through biological possibilities. No central programmer. No single organism knows the outcome. Yet new solutions continually emerge.



Evolution looks remarkably like a collective information-processing process.

A bigger question ahead

If intelligence can exist without a brain...
And perhaps even without an individual organism...
What about consciousness?

"Is consciousness something that biology produces, or something that biology organizes, concentrates or amplifies?"

— Monendra Grover (2026)

From Individual Intelligence to Collective Intelligence

In the last article, I asked a rather uncomfortable question:

Is intelligence older than brains?

We are used to associating intelligence with brains.

Humans think. Animals think. Machines can now perform tasks that look remarkably intelligent.

But bacteria can sense their environment and change their behaviour.

Plants respond to their surroundings, communicate chemically and alter their growth.

The immune system can recognize unfamiliar threats, remember previous encounters and respond differently the next time.

Slime moulds can find efficient routes through complex environments.

Ant colonies can solve problems that no individual ant could solve on its own.

So perhaps the brain is not where intelligence begins.

But this immediately leads to another question.

If intelligence can exist without a brain, can it exist without an individual?

What if intelligence is sometimes a property of a **collective**?

The individual may not be the real unit of intelligence

When we see an ant colony moving food, defending its nest or finding an efficient route, our natural instinct is to ask:

Which ant is making the decision?

The answer is: probably none of them.

There is no ant sitting in a control room saying:

“You go left. You carry the food. You defend the nest.”

And yet the colony can behave in a remarkably coordinated manner.

Each ant follows relatively simple rules.

It senses chemical signals.

It responds to the behaviour of nearby ants.

It leaves or follows trails.

It changes its behaviour depending on circumstances.

And somehow, out of all these local interactions, something much more sophisticated emerges.

The intelligence is not located in any single ant.

It emerges from the network.

This idea changes the question we should be asking.

Instead of asking only:

“Where is the intelligence?”

perhaps we should also ask:

“Between whom does the intelligence emerge?”

The whole can know something that no part knows

This is one of the most fascinating properties of biological systems.

Imagine hundreds or thousands of individuals, each possessing only limited information.

No individual has the complete picture.

Yet when they interact, the collective can arrive at a solution.

This happens in ant colonies, bee colonies, bird flocks, microbial communities and many other biological systems.

It also happens in our own bodies.

A human body is not simply a collection of independent cells.

It is an extraordinarily interconnected system.

Cells communicate.

Signals move through tissues.

Hormones travel between organs.

The nervous system coordinates distant parts of the body.

The immune system continuously monitors what is happening.

Metabolism links seemingly unrelated processes.

Information is constantly being exchanged.

And the remarkable thing is that **no single cell possesses a complete description of the organism.**

Yet the organism behaves as a coordinated whole.

Perhaps, then, the individual cell is not always the right unit at which to look for biological intelligence.

A cell is already a collective

There is an even deeper problem.

We often talk about “the cell” as though it were one thing.

But a cell is itself a vast collective.

It contains proteins, nucleic acids, membranes, molecular machines, signalling pathways, organelles and thousands of interacting chemical processes.

These components are constantly influencing one another.

A protein changes conformation.

That changes an interaction.

The interaction activates another pathway.

The pathway alters gene expression.

Gene expression changes the concentration of proteins.

Those proteins change the behaviour of the cell.

And the cycle continues.

There is no single molecule sitting at the centre issuing instructions.

The cell's behaviour emerges from **relationships among its components**.

This was one of the important questions behind the earlier articles in this series:

Can a cell compute?

Perhaps the more interesting question is now:

Can a collective compute?

And if it can compute, under some circumstances, can it also become intelligent?

The immune system: intelligence without a brain

The immune system makes this question particularly interesting.

It has no central brain.

Yet it performs tasks that look remarkably sophisticated.

It distinguishes between different molecular patterns.

It responds differently to different threats.

It can amplify some responses and suppress others.

It communicates between distant parts of the body.

And, importantly, it can retain information about previous encounters.

In other words, the immune system has something that resembles several ingredients we normally associate with intelligence:

recognition, memory, communication, adaptation and decision-making.

But where exactly is the intelligence?

Is it in an individual immune cell?

In a particular signalling molecule?

In the antibodies?

In the lymph nodes?

Or in the enormous network formed by all of these components?

The answer may be:

in the interactions.

Plants make the problem even harder

Plants push this idea further.

A plant does not have a brain.

Yet it continuously senses its environment.

Light.

Gravity.

Temperature.

Water availability.

Chemical signals.

Physical damage.

Microbial interactions.

It changes its growth accordingly.

Roots explore their surroundings.

Leaves adjust their behaviour.

Plants can alter their responses depending on previous conditions.

The plant is therefore not simply reacting mechanically to every stimulus.

It is integrating information.

But again, where is the “decision-maker”?

There isn't an obvious one.

Perhaps we are making a conceptual mistake by looking for one.

Maybe biological intelligence does not always require a central decision-maker.

From organisms to communities

Now take one more step.

If a cell can be understood as a network of interacting components, and an organism can be understood as a network of interacting cells, why should we stop at the organism?

Consider a microbial community.

Thousands or millions of microorganisms interact with one another.

They compete.

They cooperate.

They exchange molecules.

They modify their environment.

Some produce compounds that others use.

Some respond to signals released by their neighbours.

The community can therefore change its behaviour as the environment changes.

At this point, the boundaries become surprisingly difficult to define.

Where does one organism end and the collective begin?

And if the behaviour of the collective cannot be understood simply by studying one member, perhaps we need another level of biological description.

Could an ecosystem process information?

This is where the question becomes much more speculative—and, to me, much more interesting.

An ecosystem contains organisms interacting with one another and with their physical environment.

Plants influence soil.

Microorganisms transform nutrients.

Animals transport seeds.

Predators influence populations.

Climate affects organisms.

Organisms, in turn, modify their environment.

Information is therefore constantly flowing through the system.

But can we call this **information processing**?

Perhaps.

Can we call it **collective intelligence**?

That is a much stronger claim.

And I don't think we should make that claim simply because an ecosystem is complex.

Complexity by itself is not intelligence.

A hurricane is complex.

A crystal can be highly ordered.

A chemical reaction network can be extraordinarily complicated.

None of these facts, by themselves, demonstrate intelligence.

So we need criteria.

What would make a collective intelligent?

If we want to use the term “collective intelligence” scientifically, we need to be careful.

Perhaps we should look for several measurable properties.

Does the collective:

- integrate information from different sources?
- distinguish between different environmental contexts?
- modify its future behaviour based on previous experience?
- retain some form of memory?
- adapt when circumstances change?
- solve problems that individual components cannot solve?
- coordinate actions without a central controller?
- remain functional when some components fail?

These questions move us away from metaphor.

Instead of saying:

“The ecosystem is intelligent.”

we can ask:

“Does this system exhibit measurable properties that we normally associate with intelligence?”

That is a question we can investigate experimentally.

And that distinction is important.

Intelligence may exist at several levels

Perhaps the biggest change in perspective is this:

Intelligence may not have a single location.

It may exist at several levels simultaneously.

A molecule responds to its environment.

A network of molecules produces cellular behaviour.

A network of cells produces tissue behaviour.

An organism interacts with other organisms.

Groups coordinate.

Communities adapt.

Ecosystems change.

At each level, new properties can emerge.

This does not mean that every higher level is “more intelligent” than the level below it.

Rather, it suggests that **the unit of information processing can change with scale.**

The intelligence of the whole may not simply be the sum of the intelligence of its parts.

A surprising lesson from biological networks

There is something else worth noticing.

Biological networks are not designed from the top down.

Nobody designed the immune system as an engineered communication network.

Nobody programmed an ecosystem.

Nobody wrote a central algorithm for an ant colony.

And yet these systems can be remarkably robust.

They can continue functioning even when individual components fail.

They can reorganize.

They can adapt.

They can exploit local information to produce global behaviour.

This is very different from the way we normally imagine intelligent systems.

We tend to imagine a central processor receiving information, running an algorithm and producing an answer.

Biology often does something else.

The computation is distributed across the system.

Could communication itself be the key?

This brings us back to one of the themes running through my earlier work.

Perhaps we have focused too much on **information storage** and not enough on **information exchange**.

A biological system is constantly communicating with itself.

Molecules interact.

Cells signal.

Organs communicate.

Organisms interact.

Communities exchange information.

The environment itself becomes part of the information-processing process.

In some of my earlier theoretical work, I have explored much more speculative possibilities involving quantum coherence, entanglement and non-local communication in biological systems.

These ideas remain **hypotheses, not established biological mechanisms**.

There is currently no experimental basis for claiming that biological organisms routinely use the kind of long-range quantum communication sometimes proposed in these models.

But I think such hypotheses are useful if they lead us to ask better questions.

For example:

Are there forms of biological coordination that our current models do not adequately capture?

And, importantly:

How could we experimentally prove them wrong?

That last question matters.

A scientific hypothesis becomes interesting not merely because it is imaginative, but because it can be tested.

From “group behaviour” to collective intelligence

There is an important distinction here.

A group behaving together is not necessarily intelligent.

Thousands of people walking through a railway station produce coordinated patterns.

But that does not automatically mean the crowd is thinking.

Collective intelligence requires something more.

There must be some combination of:

information integration + interaction + adaptation + memory + context-sensitive behaviour.

And perhaps, most importantly, the collective should be capable of doing something that its individual members cannot accomplish alone.

That is where the concept becomes scientifically interesting.

Evolution itself looks surprisingly collective

There is another connection that I find particularly intriguing.

Evolution is often described as a process acting on individual organisms.

But evolution also depends on interactions.

Genes interact with genes.

Organisms interact with organisms.

Organisms interact with environments.

Communities modify selection pressures.

Cooperation can create new possibilities.

Competition can drive adaptation.

The environment is not simply a background.

It participates in the process.

Seen this way, evolution begins to resemble a vast distributed search through biological possibilities.

There is no central programmer.

No single organism knows where the process is going.

Yet new solutions continually emerge.

This does not mean that evolution “thinks.”

But it does suggest that **intelligence-like properties can emerge from distributed processes without a central intelligence directing them.**

What does this mean for artificial intelligence?

This question is particularly relevant today.

We are building increasingly powerful artificial systems by giving machines access to enormous amounts of information and increasingly sophisticated computational architectures.

But biology suggests another possibility.

Perhaps intelligence does not require a single powerful processor.

Perhaps intelligence can emerge from:

many simple units + communication + feedback + memory + adaptation.

That is a very different design philosophy.

And it raises an intriguing possibility for future AI:

Instead of trying only to build a bigger artificial “brain,” perhaps we should also learn how to build **intelligent collectives**.

Not one system doing everything.

Many systems interacting, learning and adapting together.

In some ways, biology has been exploring this idea for billions of years.

Perhaps the boundary between individual and collective is artificial

The more deeply we look at biology, the harder it becomes to draw a sharp boundary around the “individual.”

A human being is made of trillions of cells.

Those cells communicate continuously.

We depend on microorganisms living with us.

Our behaviour influences other organisms.

Other organisms influence us.

We are therefore not isolated information-processing entities.

We are **networks embedded within larger networks**.

Perhaps individuality is real at one level and incomplete at another.

A cell is an individual.

But it is also part of a tissue.

An organism is an individual.

But it is also part of a population.

A population is an individual unit from one perspective, but part of an ecosystem from another.

The hierarchy never really ends.

A new picture of intelligence

So where does this leave us?

Perhaps intelligence is not a thing that suddenly appeared when the first nervous system evolved.

Perhaps it is a property that can emerge whenever matter becomes sufficiently organized to:

sense → integrate → remember → adapt → act.

Sometimes this happens inside a brain.

Sometimes inside a cell.

Sometimes across a network of cells.

Sometimes across many organisms.

And perhaps, under the right conditions, at still larger scales.

We should not assume that every complex system is intelligent.

But neither should we assume that intelligence must have a brain.

That was the lesson of Article 7.

Article 8 takes the idea one step further:

Perhaps intelligence does not even have to belong to an individual.

It may sometimes belong to the **relationships between individuals.**

And that changes the question completely.

Instead of asking only:

“Is this organism intelligent?”

we may eventually need to ask:

“Is the system formed by these organisms capable of processing information in ways that none of them can achieve alone?”

That is a question we can begin to test.

And perhaps it will force us to rethink not only intelligence, but also what we mean by an **individual**, an **organism**, and even a **living system**.

And this leads us to the next question

If intelligence can exist without a brain...

And perhaps even without an individual organism...

then another question becomes almost unavoidable:

What about consciousness?

Is consciousness something that biology eventually produced through sufficiently complex brains?

Or could consciousness be something more fundamental—something that biology organizes, concentrates or amplifies rather than creates from nothing?

This takes us into much more uncertain territory.

And here, I want to be especially careful.

The scientific evidence for the ideas I will discuss next is far less established than the biology discussed in the earlier articles.

But uncertainty is not a reason to stop asking questions.

It is a reason to ask them more carefully.

Perhaps the next step in rethinking biology is to ask whether consciousness is a product of biology—or a property of reality itself.

That is where our journey goes next.

A note on evidence

As in the earlier articles in this series, I have deliberately separated three levels of discussion:

Established science: distributed biological signalling, cellular communication, immune memory, collective behaviour and emergence.

Scientific extrapolation: the possibility that some of these properties can reasonably be described as forms of collective information processing or intelligence.

Speculative hypothesis: ideas involving deeper quantum mechanisms, non-local biological communication or consciousness as a fundamental property of reality.

The purpose of the *Rethinking Biology* series is not to present speculation as fact.

It is to ask:

What happens when we take established observations seriously and follow their implications to the edge of what we currently understand?

And then ask the most important scientific question of all:

How could we test it?